claude-windows / 42211f80-2169-4025-ad9d-99c1aa29ff70

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\subagents\workflows\wf_6237b110-d58\agent-a22dd3587d7a3c9b6.jsonl`
- SHA-256: `4205585b907ad7b5bb2a577a5a22af8807930402ceded5fd1a20ba4c64e4fa6b`
- Source modified: `2026-06-06T17:31:33+00:00`
- Imported at: `2026-07-05T16:48:28+00:00`
- project: `wf_6237b110-d58`
- session_id: `42211f80-2169-4025-ad9d-99c1aa29ff70`

Transcript

1. user (2026-06-06T17:30:14.119Z)

Source Extractor

```
Research question: "Research alternative methods to automatically detect rally start/end
segments (temporal segmentation) in racket-sport video ? especially badminton ? for a SELF-
HOSTED, LOCAL-FIRST pipeline, and assess whether a cloud-VLM teacher ? local-model distillation
is the best approach or whether other methods are competitive or complementary. Keep it focused
("mini" research): prioritize actionable, commercially-licensable, locally-runnable approaches
over exotic SOTA needing huge compute or restrictive licenses.

OUR CONTEXT / CONSTRAINTS (ground every recommendation in these):
- Self-hosted on a single consumer GPU (RTX 2070 Super), Windows + WSL2. We want LOCAL-ONLY
inference at runtime ? privacy lever: user video never leaves the device. It's a commercial
product, so PERMISSIVE / commercial-friendly licenses are REQUIRED (avoid AGPL and non-
commercial-only weights/datasets).
- Current substrate: WASB (MIT) shuttle-trajectory detector ? per-frame (x, y, visibility). We
derive rally windows via velocity/merge-gap/min-duration thresholds plus a net-crossing gate. We
ALSO have a permissive person detector available (torchvision Faster R-CNN + supervision
ByteTrack) and the video's AUDIO track.
- Current best approach: use paid Gemini (cloud VLM, full-context) OFFLINE as a TEACHER to label
rallies, then DISTILL into a local model (logistic regression on shuttle-trajectory features).
Findings so far: threshold-tuning ceilings around F1 ? 0.44 and overfits with few videos; the
learned model is data-starved at n=2 labeled videos; WASB candidate RECALL is a real bottleneck
(its windows sometimes don't align with true rallies at IoU 0.5).
- Inputs we can exploit locally: video frames, the shuttle trajectory, player bounding boxes /
pose (if we add a permissive pose model), and the AUDIO track (shuttle "thwack" hits, crowd
noise, umpire/score calls).

SPECIFICALLY COVER:
1. Method families for rally/point segmentation in racket sports (badminton, tennis, table
tennis, pickleball): audio-based shot/hit detection; temporal action localization (TAL) deep
models; pose/player-stance + court-homography state machines (e.g., "serve-ready stance"
gating); classical change-point detection / HMM on the shuttle trajectory; scoreboard / score-
overlay OCR for point boundaries; self-supervised / weakly-supervised approaches; and VLM
teacher?student distillation.
2. For EACH family: how it works, the labels/data it needs, expected accuracy, local-inference
COMPUTE cost, and LICENSE / commercial viability of any models, datasets, or weights involved.
3. Badminton rally-segmentation datasets (e.g., ShuttleSet, TrackNet/TrackNetV2 data) and their
licenses / commercial usability.
4. A RANKED recommendation for our local-first, permissive, single-GPU setup: is Gemini-teacher
? local-distillation the best, or are some alternatives competitive or complementary ?
especially AUDIO-based hit detection and pose/court-geometry state machines, which we can run
fully locally?
5. Concrete, cheap next experiments we could run with what we already have (shuttle trajectory +
person detector + audio)."
```

Fetch and extract key claims from this source:
URL: https://arxiv.org/pdf/2308.12199
Title: Towards Real-Time Analysis of Broadcast Badminton Videos (arXiv 2308.12199)
Found via: broad/primary search

Task
1. Use WebFetch to retrieve the page content.
2. Assess source quality: primary research/institution? secondary reporting? blog/opinion? forum? unreliable?
3. Extract 2-5 FALSIFIABLE claims that bear on the research question. Each claim must:
   - be a concrete, checkable statement (not vague generalities)
   - include a direct quote from the source as support
   - be rated central/supporting/tangential to the research question
4. Note publish date if available.

If the fetch fails or the page is irrelevant/paywalled, return claims: [] and sourceQuality: "unreliable".

Structured output only.

2. assistant (2026-06-06T17:30:17.750Z)

I'll fetch the source and extract key claims relevant to the research question.

3. user (2026-06-06T17:30:32.508Z)

maxContentLength size of 10485760 exceeded

4. assistant (2026-06-06T17:30:35.009Z)

The PDF is too large to fetch directly. Let me try the abstract page and the HTML version.

5. user (2026-06-06T17:30:42.270Z)

Analysis of arXiv:2308.12199

Publication Details
- **Title:** Towards Real-Time Analysis of Broadcast Badminton Videos
- **Date:** Submitted August 23, 2023
- **Authors:** Nitin Nilesh, Tushar Sharma, Anurag Ghosh, C. V. Jawahar
- **Institution:** Not explicitly stated in provided content

Relevant Claims Found

Player Tracking & Movement Analysis:
The paper proposes "an end-to-end framework for player movement analysis for badminton matches on live broadcast match videos" using only visual inputs. The authors calculate "on-court distance covered by both the players" and demonstrate "average speed with which they move on the court."

Real-Time Performance:
The framework "was successfully used to analyze live broadcast matches in real-time during the Premier Badminton League 2019 (PBL 2019)."

Claims NOT Found in Provided Content

The abstract and metadata provided do **not** contain information regarding:
1. Rally/point/stroke segmentation methods
2. Audio-based hit detection
3. Pose/player-stance or court-homography state machines
4. Shuttle trajectory tracking
5. Specific datasets and licenses

6. Accuracy/F1 scores
7. Detailed compute costs
8. Self-supervised/weakly-supervised learning approaches

Note: Full paper content would be required to extract the technical details you requested.

6. user (2026-06-06T17:30:44.654Z)

maxContentLength size of 10485760 exceeded

7. assistant (2026-06-06T17:30:47.467Z)

I have the abstract details. Let me try to get the full text via the arxiv HTML (ar5iv) version to extract technical details on methods, datasets, and accuracy numbers.

8. user (2026-06-06T17:30:50.527Z)

REDIRECT DETECTED: The URL redirects to a different host.

Original URL: https://ar5iv.org/abs/2308.12199
Redirect URL: https://ar5iv.labs.arxiv.org/abs/2308.12199
Status: 301 Moved Permanently

To complete your request, I need to fetch content from the redirected URL. Please use WebFetch again with these parameters:
- url: "https://ar5iv.labs.arxiv.org/abs/2308.12199"
- prompt: "Extract technical details about: (1) how players are detected/tracked, (2) court detection / homography, (3) any rally or point or stroke segmentation, (4) shuttle tracking, (5) datasets used and their licenses/source, (6) accuracy/precision/F1 numbers reported, (7) real-time / FPS / compute cost, (8) any self-supervised/weakly-supervised learning. Quote exact sentences with numbers."

9. user (2026-06-06T17:31:06.325Z)

Technical Details from "Towards Real-Time Analysis of Broadcast Badminton Videos"

(1) Player Detection/Tracking

"A YOLOv3 network is finetuned for these two classes" (PlayerTop and PlayerBottom). The network uses transfer learning, first pre-trained on PASCAL-VOC dataset, then fine-tuned on their custom dataset with 3000 annotated frames. Player position is tracked as "the center bottom point of the player bounding box."

(2) Court Detection/Homography

"Using homography techniques, we map the player locations from broadcast camera view" to top-view court coordinates. The method "find[s] the court lines from the video frame using Hough line finding method and get[s] the position of court corners." They compute homography between broadcast view and "top view four court corners to get a mapping from camera coordinates to top view space."

(3) Rally/Point Segmentation

"Rally frames are usually always shot from the broadcast camera view whereas the non-rally frames can include close-ups." A finetuned ResNet-18 classifier distinguishes these. "A rally is at least 3 seconds long and there is a minimum of 3 seconds gap between two rallies."

(4) Shuttle Tracking

No shuttle/shuttlecock tracking is performed in this work.

(5) Datasets

Two annotated datasets from "Badminton World Federation (BWF) channel on YouTube"?10 singles matches and 10 doubles matches with "22 hours 16 minutes data." A third dataset consists of "25 rallies" of amateur players with "distance trackers...accurate up to 1 meter."

(6) Accuracy Numbers

Point Segmentation (CNN vs. HOG+SVM):
- CNN: "F1-Score 97.23, Precision 98.06, Recall 92.34"
- HOG+SVM: "F1-Score 95.44, Precision 97.83, Recall 91.02"

Player Localization (YOLOv3):
- "mAP@0.5 Player Top 98.07, mAP@0.5 Player Bottom 98.13"

Distance Calculation (25 rallies ground truth):
- Mean error Player Top: "4.5 meters," Player Bottom: "4.73 meters"
- Max error: "6 meters" (Player Top), Min: "1 meter" (Player Bottom)

(7) Real-Time Performance

"Match Latency: CNN 16.75 minutes" (vs. HOG+SVM 57.24 minutes for a full match). "Rally Latency: YOLOv3 0.14 seconds" per frame. "One rally is 10 to 12 seconds long...250 to 300 rally frames." They process "every 3rd rally frame" for real-time deployment at PBL 2019.

(8) Self/Weakly-Supervised Learning

No self-supervised or weakly-supervised learning approaches are employed; the method relies on manual annotation for training.

10. assistant (2026-06-06T17:31:14.058Z)

I now have detailed technical content. Let me verify the key rally-segmentation numbers and the rally definition by fetching the relevant section once more to confirm the "rally vs non-rally frame classifier" framing is accurate.

The extraction is consistent and detailed. I have enough to produce falsifiable claims. The most research-relevant finding is the rally/non-rally frame classifier (a shot-type/broadcast-view classifier approach to point segmentation that runs locally without shuttle tracking), the homography/court-line state, and the heuristic rally definition (3s min duration, 3s min gap) ? which directly parallels the user's own merge-gap/min-duration approach.

11. user (2026-06-06T17:31:24.846Z)

Structured output provided successfully

12. assistant (2026-06-06T17:31:33.448Z)

I have completed the source extraction and returned the structured output.

Summary of what I found in arXiv:2308.12199 ("Towards Real-Time Analysis of Broadcast Badminton Videos," Nilesh/Sharma/Ghosh/Jawahar, IIIT-style group, submitted 2023-08-23 ? a primary research paper):

The most research-relevant finding is that this paper segments points/rallies using a **per-frame CNN image classifier** (fine-tuned ResNet-18) that distinguishes broadcast-camera rally frames from non-rally frames (replays/close-ups), hitting F1 97.23 ? entirely without shuttle tracking. That is a genuinely different, fully-local method family than the user's WASB-trajectory approach, and notably it relies on the broadcast-production cut structure (camera angle changes), which is a strong signal in produced footage but may not transfer to single-camera amateur recordings.

Two caveats worth flagging for the parent synthesis: (1) it uses the *same* heuristic temporal constraints the user already employs (?3s rally duration, ?3s inter-rally gap), so it validates that approach rather than superseding it; and (2) it is fully supervised on manual annotations with no audio, pose-state-machine, self-supervised, or VLM-distillation component ? so it is a complementary visual-segmentation baseline, not a teacher-distillation alternative.